

Exercise Solution

- 1.1 Convert the following binary numbers to the equivalent octal numbers.

$$(b) (101110.0111)_2 = (56.34)_8$$

- 1.2 Convert the following binary numbers to the equivalent hexadecimal numbers.

$$(c) (10101.1101)_2 = (15.D)_{16}$$

- 1.3 Convert the following octal numbers to the equivalent binary numbers.

$$(b) (67.23)_8 = (110111.010011)_2$$

- 1.4 Convert the following hexadecimal numbers to the equivalent binary numbers.

$$(c) (25C.38)_{16} = (1001011100.00111)_2$$

- 1.5 Convert the following hexadecimal numbers to the equivalent octal numbers.

$$(e) (8E36.7A)_{16} = (107066.364)_8$$

- 1.6 Convert the following binary numbers to the equivalent decimal numbers.

$$(c) (10101.1101)_2 = (21.8125)_{10}$$

- 1.7 Convert the following octal numbers to the equivalent decimal numbers.

$$(c) (66.24)_8 = (54.3125)_{10}$$

- 1.8 Convert the following hexadecimal numbers to the equivalent decimal numbers.

$$(c) (15D.38)_{16} = (349.21875)_{10}$$

- 1.9 Convert the following decimal numbers to the equivalent binary numbers.

$$(c) (133)_{10} = (10000101)_2$$

- 1.10 Convert the following decimal numbers to the equivalent octal numbers.

$$(c) (9713)_{10} = (22761)_8$$

- 1.11 Convert the following decimal numbers to the equivalent hexadecimal numbers.

$$(c) (65112)_{10} = (FE58)_{16}$$

1.12 Write the Signed-Magnitude, the Ones' Complement, and the Two's Complement of the following binary numbers.

$$(a) (+1011)_2 = [01011]_{SM} = [01011]_{OC} = [01011]_{TC}$$

$$(b) (-1011)_2 = [11011]_{SM} = [10100]_{OC} = [10101]_{TC}$$

$$(c) (+0.0011)_2 = [0.0011]_{SM} = [0.0011]_{OC} = [0.0011]_{TC}$$

$$(d) (-0.0011)_2 = [1.0011]_{SM} = [1.1100]_{OC} = [1.1101]_{TC}$$

1.13 Calculate the addition of the following binary numbers in the two's complement.

$$X_1 = (-0110110)_2$$

$$X_2 = (-0011110)_2$$

$$X_1 = (-0110110)_2 = [10110110]_{SM} = [11001001]_{OC} = [11001010]_{TC}$$

$$X_2 = (-0011110)_2 = [10011110]_{SM} = [11100001]_{OC} = [11100010]_{TC}$$

$$\begin{aligned} X_1 + X_2 &= [11001010]_{TC} + [11100010]_{TC} = [10101100]_{TC} \\ &= [11010100]_{SM} = (-1010100)_2 \end{aligned}$$

$$\begin{array}{r} 11001010 \\ + \quad 11100010 \\ \hline \mathbf{1}10101100 \end{array}$$

Drop the carry on the sign bit.